

Honours Analysis 1 Tutorials (Sec. 003)

Edward Chernysh

Please note that this document has not been proofread and may contain typos. Feel free to see or email me if you have any questions.

REMINDER: There are people available at the Math helpdesk (Burnside 911) to help answer questions you might have. Please note that my office hours on Thursdays have changed to 3:30pm - 4:30pm. However, my office hours on September 13th will be from 3:00pm-4:00pm.

1 Second Tutorial

We would first like to recall some notions regarding functions. Let X and Y be sets, a *bijection* from X to Y is a function $f : X \rightarrow Y$ satisfying the following properties:

- (1) f is *injective*, i.e. $f(x) = f(x')$ implies $x = x'$;
- (2) f is *surjective*, i.e. for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.

Put otherwise, a function $f : X \rightarrow Y$ is a bijection *if and only if* for every $y \in Y$ there exists a *unique* point $x \in X$ such that $f(x) = y$.

Example 1. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ is not a bijection because it is neither injective nor surjective. Certainly, injectivity fails because $f(-1) = 1 = f(1)$. Surjectivity does not hold because there does not exist $x \in \mathbb{R}$ such that $f(x) = -1$.

We must be careful, as a function that is not surjective can become surjective if we restrict its range. Consider the function

$$f : \mathbb{R} \rightarrow [0, \infty), \quad f(x) = x^2.$$

This is the same function as in the previous example, but with a “smaller” range. Although the function is still not injective ($f(-1) = f(1)$), it is surjective! Indeed, given $y \in [0, \infty)$, we have that $f(\sqrt{y}) = y$.

Problem 1. Let X and Y be sets and $f : X \rightarrow Y$ a bijection. Let $f^{-1} : Y \rightarrow X$ be the inverse function of f . Then, f^{-1} is a bijection $Y \rightarrow X$.

Proof. We first check that f^{-1} is surjective. This amounts to showing that for every $x \in X$, there exists $y \in Y$ with the property that $f^{-1}(y) = x$. Given $x \in X$, we simply take $y = f(x) \in Y$. Then,

$$f^{-1}(y) = f^{-1}(f(x)) = x.$$

Therefore, f is surjective. To prove injectivity we assume that $f^{-1}(y) = f^{-1}(y')$ for some $y, y' \in Y$. We want to show that this forces $y = y'$. To this end, we “apply” the function f to both sides of the equality $f^{-1}(y) = f^{-1}(y')$. Doing so gives

$$f(f^{-1}(y)) = f(f^{-1}(y')) \implies y = y'.$$

Hence, f is injective. □

Problem 2. Let A, B and C be sets, and let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. Prove the following:

- (1) if f and g are injective, then so is $g \circ f$;
- (2) if $g \circ f$ is injective, then so is f ;
- (3) if f and g are surjective, then so is $g \circ f$;
- (4) if $g \circ f$ is surjective, then so is g ;
- (5) give an example where f is not surjective and g is not injective, but $g \circ f$ is bijective;

Proof.

- (1) Let $a, a' \in A$ and suppose that $(g \circ f)(a) = (g \circ f)(a')$; our goal is to prove that $a = a'$. Now, this equation is equivalent to $g(f(a)) = g(f(a'))$. Because g is injective, this implies $f(a) = f(a')$. But, f is also injective! Consequently, it follows that $a = a'$.

- (2) Suppose that $g \circ f$ is injective and let $a, a' \in A$ be such that $f(a) = f(a')$. We want to deduce that $a = a'$. Let us now apply the function g to both sides of the equality $f(a) = f(a')$. Doing so gives

$$(g \circ f)(a) = g(f(a)) = g(f(a')) = (g \circ f)(a').$$

Because $g \circ f$ is injective, it follows that $a = a'$. We conclude that f is injective.

- (3) Let f and g both be surjective. For any point $c \in C$, we can choose $b \in B$ such that $g(b) = c$. Since f is surjective, there exists $a \in A$ with the property that $f(a) = b$. Then,

$$(g \circ f)(a) = g(f(a)) = g(b) = c.$$

- (4) If $g \circ f$ is surjective, given any point $c \in C$ there exists $a \in A$ such that $(g \circ f)(a) = c$. This is simply the statement that $g(f(a)) = c$. Taking $b := f(a) \in B$, we see that $g(b) = c$. Hence, g is surjective.
- (5) Let $\mathbb{E}$ denote the set of all even natural numbers $\{2, 4, 6, \dots\}$. Consider the functions:

$$\begin{aligned} f : \mathbb{E} &\rightarrow \mathbb{N}, & f(n) &= n, \\ g : \mathbb{N} &\rightarrow \mathbb{E}, & g(n) &= \begin{cases} n & \text{if } n \text{ is even} \\ 2n & \text{if } n \text{ is odd.} \end{cases} \end{aligned}$$

Clearly, f is injective but not surjective. Because $g(1) = g(2) = 2$, we see that g is not injective. But, $g \circ f$ is simply the identity map

$$g \circ f : \mathbb{E} \rightarrow \mathbb{E}, \quad (g \circ f)(n) = n$$

which is a bijection (check this yourself).

□

The following exercise is to be expected, and the result is highly intuitive.

Problem 3. *Composition to the right or left does not affect the injectivity or surjectivity of a function. Namely, the following properties hold:*

- (1) Let A, B and C be sets. Suppose that $f : A \rightarrow B$ is a function and $g : B \rightarrow C$ is a bijection. Then, $g \circ f$ is injective whenever f is. Moreover, $g \circ f$ is surjective whenever f is.
- (2) Let A, B and C be sets. Suppose that $f : A \rightarrow B$ is a bijection and $g : B \rightarrow C$ a general function. Then, $g \circ f$ is injective whenever g is. Furthermore, $g \circ f$ is surjective if g is.

Proof. Let us prove (1). First, note that g is both injective and surjective. If f is injective, then $g \circ f$ is injective by part (1) of the previous problem. If f is surjective, then so is $g \circ f$ (by part (3) of the last problem).

Part (2) is verified in a similar way. We leave the details as an exercise to the reader. $\square$

From this, we can deduce the following.

Corollary 1.1. Let A, B and C be sets and $f : A \rightarrow B, g : B \rightarrow C$ functions. If any two of f, g and $g \circ f$ are bijective, then so is the third.

Proof. If f and g are bijective, then parts (1) and (3) of Problem 2 tell us that $g \circ f$ is also a bijection. Suppose that f and $g \circ f$ are bijections. Then, f has an inverse function f^{-1} . By Problem 1, this inverse is also a bijection. Since composition does not affect either injectivity or surjectivity (by the last problem), we see that

$$g = g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1}$$

is bijective. Similarly, f can be shown to be bijective whenever g and $g \circ f$ are. $\square$

Problem 4. Let X and Y be sets and $f : X \rightarrow Y$ a function. Show that f is injective if and only if there exists a function (called a left-inverse) $g : Y \rightarrow X$ such that

$$g \circ f : X \rightarrow X$$

satisfies $(g \circ f)(x) = x$ for all $x \in X$.

Proof. Suppose that f is injective. If $y \in f(X)$, then there exists a unique point $x \in X$ such that $y = f(x)$. Let us call this *unique* point $f^{-1}(y)$. Define a function

$$g : Y \rightarrow X, \quad g(y) := \begin{cases} f^{-1}(y), & \text{if } y \in f(X), \\ x_0, & \text{if } y \notin f(X). \end{cases}$$

Here, $x_0 \in X$ is **any** point. Let $x \in X$ be given, we want to verify that

$$(g \circ f)(x) = g(f(x)) = x.$$

Clearly, $y = f(x)$ belongs to $f(X)$. By definition of the function g ,

$$g(f(x)) = g(y) = f^{-1}(y) = x.$$

Conversely, suppose that such a function g exists. The composition $g \circ f$ is then equal to the map $h(x) = x$ on X . Thus, $g \circ f$ is an injection $X \rightarrow X$. By part (2) of Problem 2, we see that f is injective. $\square$

1.1 More Induction

The following problems are meant to serve as extra examples of mathematical induction. As an added bonus, we will be proving inequalities that you will probably find to be useful throughout your entire mathematical career. At the very least, the first inequality should help you in Math 248.

Let us recall Problem 4 in your first assignment, which states that

$$\left(\sum_{j=1}^n a_j b_j \right)^2 \leq \left(\sum_{j=1}^n a_j^2 \right) \left(\sum_{j=1}^n b_j^2 \right) \quad (1.1)$$

for all $n \in \mathbb{N}$ and all $a_j, b_j \in \mathbb{R}$. In fact, this implies that

$$\sum_{j=1}^n a_j b_j \leq \left(\sum_{j=1}^n a_j^2 \right)^{1/2} \left(\sum_{j=1}^n b_j^2 \right)^{1/2}. \quad (1.2)$$

Let $x = (x_1, \dots, x_n)$ be a point in $\mathbb{R}^n$ for $n \geq 1$. The *norm* of the point x is defined via the equation

$$\|x\| := (x_1^2 + \dots + x_n^2)^{1/2} \geq 0 \quad (1.3)$$

If $y = (y_1, \dots, y_n)$ is another point in $\mathbb{R}^n$, we define the *dot product* of x and y according to the formula

$$x \cdot y = x_1 y_1 + \dots + x_n y_n. \quad (1.4)$$

This is an example of what is often called an *inner product*. Returning to the equation above, we see that

$$x \cdot x = x_1^2 + \dots + x_n^2 \geq 0.$$

Note that $\|x\|^2 = x \cdot x$ whence $\|x\| = \sqrt{x \cdot x}$. This relationship means that the norm $\|x\|$ is *induced* by the dot product (this topic will be explored in Honours Analysis 4).

Problem 5. Let $n \geq 1$ and $x, y \in \mathbb{R}^n$. Then, $\|x + y\| \leq \|x\| + \|y\|$.

Proof. By our earlier observations,

$$\begin{aligned} \|x + y\|^2 &= (x + y) \cdot (x + y) = \sum_{j=1}^n (x_j + y_j)^2 \\ &= \sum_{j=1}^n x_j^2 + 2 \sum_{j=1}^n x_j y_j + \sum_{j=1}^n y_j^2 \\ &= \|x\|^2 + 2 \sum_{j=1}^n x_j y_j + \|y\|^2. \end{aligned}$$

Invoking (1.2), we get

$$\begin{aligned} \|x + y\|^2 &\leq \|x\|^2 + 2 \left(\sum_{j=1}^n x_j^2 \right)^{1/2} \left(\sum_{j=1}^n y_j^2 \right)^{1/2} + \|y\|^2 \\ &= \|x\|^2 + 2 \|x\| \|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2. \end{aligned}$$

Taking square roots gives us the desired inequality. $\square$

Theorem 1.2. For each $n \geq 1$, and all $x_1, \dots, x_n > 0$, there holds

$$\frac{x_1 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}. \quad (1.5)$$

Remark 1.1. The quantity on the left hand side of (1.5) is called the *arithmetic mean* of $(x_1, \dots, x_n)$. The right hand term is instead called the *geometric mean* of $(x_1, \dots, x_n)$.

To prove this result, we will first require a lemma.

Lemma 1. Let $n \in \mathbb{N}$ and $x_1, \dots, x_n > 0$. If $x_1 \cdots x_n = 1$, then

$$x_1 + \dots + x_n \geq n.$$

Proof. We prove this inequality by induction on $n \geq 1$. Luckily for us, the base case with $n = 1$ is obvious.

Inductive Step. Suppose that the claim holds for an $n \geq 1$. We will prove that the statement holds for $n + 1$. Thus, our inductive hypothesis is the statement:

$$“(\forall x_1, \dots, x_n > 0), x_1 \cdots x_n = 1 \implies x_1 + \cdots + x_n \geq n.”$$

Let $x_1, \dots, x_n, x_{n+1} > 0$ be a sequence of real numbers such that $x_1 \cdots x_n x_{n+1} = 1$. We will show that

$$x_1 + \cdots + x_n + x_{n+1} \geq n + 1.$$

If every $x_j = 1$, then the statement is obvious. Thus, there must exist indices $1 \leq i, j \leq n + 1$ such that $x_i > 1$ and $x_j < 1$. After a relabeling, we can assume that $x_1 < 1$ and $x_{n+1} > 1$. Because $(x_{n+1} - 1)(1 - x_1) > 0$, we must have

$$x_1 + x_{n+1} > 1 + x_1 x_{n+1}. \quad (1.6)$$

Define $y := x_1 x_{n+1}$, the inequality above can then be written as

$$x_1 + x_{n+1} > 1 + y. \quad (1.7)$$

Note that $y x_2 \cdots x_n = 1$. Since $\{y, x_2, \dots, x_n\}$ consists of n -numbers, our inductive hypothesis implies

$$y + x_2 + \cdots + x_n \geq n. \quad (1.8)$$

Using this together with (1.7) yields

$$\begin{aligned} x_1 + x_2 + \cdots + x_n + x_{n+1} &= (x_1 + x_{n+1}) + x_2 + \cdots + x_n \\ &> (1 + y) + x_2 + \cdots + x_n \quad (\text{by (1.7)}) \\ &= 1 + y + x_2 + \cdots + x_n \\ &\geq 1 + n \quad (\text{by (1.8)}) \\ &= n + 1. \end{aligned}$$

This completes the inductive step. □

We are finally ready to give a nice proof of Theorem 1.2.

Proof of Theorem 1.2. Given $x_1, \dots, x_n > 0$ let us put $A := (x_1 \cdots x_n)^{\frac{1}{n}} > 0$. For each $1 \leq j \leq n$ define

$$a_j := \frac{x_j}{A} > 0$$

Then, $a_1 \cdots a_n = 1$. By the previous lemma, we get that

$$a_1 + \cdots + a_n \geq n.$$

Or, rather, that

$$\frac{x_1 + \cdots + x_n}{A} = \frac{x_1 + \cdots + x_n}{(x_1 \cdots x_n)^{\frac{1}{n}}} \geq n.$$

Therefore,

$$\frac{x_1 + \cdots + x_n}{n} \geq (x_1 \cdots x_n)^{\frac{1}{n}}$$

as was asserted. □

Email: edward.chernysh@mail.mcgill.ca

Office: BURN 1122

Office Hours: Wed. 11:00am - 12:00pm & Thurs. 3:30pm - 4:30pm.